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A SIEM-Integrated Cybersecurity Prototype for Insider Threat Anomaly Detection Using Enterprise Logs and Behavioural Biometrics

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Abstract

Insider threats remain a serious concern for organisations in both public and private sectors. Detecting anomalous behaviour in enterprise environments is critical for preventing insider incidents. While many prior studies demonstrate promising results using deep learning on offline datasets, few address real-time operationalisation or calibrated alert control within a Security Information and Event Management (SIEM) workflow. This paper presents a SIEM-integrated prototype that fuses the Computer Emergency Response Team Insider Threat Test Dataset (CERT) enterprise logs (Logon, Device, HTTP, and Email) with behavioural biometrics from the Balabit mouse dynamics dataset. Per-modality one-dimensional convolutional neural network (1D CNN) branches are trained independently using imbalance-aware strategies, including downsampling, class weighting, and focal loss. A unified 20 × N feature schema ensures train–serve parity and consistent feature validation during live inference. Post-training calibration using Platt and isotonic regression enables analyst-controlled threshold tuning and stable alert budgeting inside the SIEM. The models are deployed in Splunk's Machine Learning Toolkit (MLTK), where dashboards visualise anomaly timelines, risky users or hosts, and cross-stream overlaps. Evaluation emphasises operational performance, precision–recall balance, calibration stability, and throughput rather than headline accuracy. Results show calibrated, controllable alert volumes: for Device, precision ≈0.70 at recall ≈0.30 (PR-AUC = 0.468, ROC-AUC = 0.949); for Logon, ROC-AUC = 0.936 with an ultra-low false-positive rate at a conservative threshold. Batch CPU inference sustains ≈70.5 k windows/s, confirming real-time feasibility. This study's main contribution is to demonstrate a calibrated, multi-modal CNN framework that integrates directly within a live SIEM pipeline. It provides a reproducible path from offline anomaly detection research to Security Operations Centre (SOC)-ready deployment, bridging the gap between academic models and operational Cybersecurity practice.

Keywords: insider threat; anomaly detection; SIEM; Splunk MLTK; convolutional neural networks; behavioural biometrics; calibration; precision–recall; class imbalance; 1D CNN

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1. Introduction

Cybersecurity research is essential as modern organisations depend on connected systems for daily operations and national infrastructure protection. Spotting hostile activity in time is a persistent challenge for both business and government contexts [1,2]. Security teams collect vast streams of enterprise logs (Logon, Email, HTTP, Device, File) through

Security Information and Event Management (SIEM) platforms to reconstruct events and detect abuse, but static rules and signature filters struggle with evolving behaviours and produce high alert volumes [2,3]. Insider threats are especially problematic because trusted users can misuse legitimate access across many channels, blending abnormal actions with routine work and evading rule-based controls [3,4].

Anomalies are data points or sequences that do not fit the usual pattern of the whole dataset. In security logs, anomalies can appear as follows: (i) point anomalies (a single unusual event), (ii) contextual anomalies (an event that is only unusual in a specific context, such as time of day or host), and (iii) collective anomalies (a sequence that is normal per event but abnormal as a group). Choosing a detector depends on data type, latency needs, and label availability; deep learning is often used because it can learn useful features directly from data and support real-time scoring.

Anomaly detection on enterprise logs seeks to learn what is "normal" for users and assets, then flag departures. Classical supervised machine learning approaches (SVM, Random Forest, XGBoost) have been applied to the CMU-CERT Insider Threat Test Dataset (CERT), log data with engineered features, and show promise, yet they can be sensitive to class imbalance, feature leakage, and non-stationary behaviour in production [3]. Unsupervised and one-class methods (OC-SVM, Isolation Forest) avoid the need for labelled attacks and are attractive under rare-event conditions, but often rely on hand-crafted summaries and global thresholds that do not adapt well to user-specific drift [5,6].

Between fully supervised and unsupervised approaches lie self-/semi-supervised approaches—sequence models trained with proxy tasks (e.g., masked event prediction) learn bidirectional context without labelled anomalies and reduce dependence on manual features [7]. Regardless of method, evaluation matters: user- and time-disjoint splits and reporting precision–recall under a fixed alert budget avoid optimistic results from random mixes and better reflect Security Operations Centre (SOC) reality [7,8].

Deep learning has been adopted to capture temporal context and complex patterns in logs. CNN/RNN hybrids and attention mechanisms have improved over traditional pipelines by learning features directly from sequences of activities, and recent self-supervised Transformers (BERT for logs) model bidirectional context without labelled anomalies [1]. These advances are complemented by behavioural biometrics—keystroke dynamics and mouse usage—which add a human-pattern signal for continuous authentication and impersonation resistance in enterprise settings [9,10]. Public datasets and benchmarks make this tractable—the CMU keystroke corpus and related studies for typing rhythms, the Balabit and Bogazici mouse-dynamics sets for cursor behaviour, and the CERT insider-threat test datasets for multi-source enterprise activity

Despite progress, three practical gaps remain for operations. First, many evaluations emphasise accuracy/ROC on random splits rather than user/time-disjoint protocols and precision–recall trade-offs at analyst-friendly alert rates, which is what matters in a SOC workflow [3]. Second, deployment inside SIEM requires calibrated scores, adaptive thresholds, and real-time scoring paths so analysts can act on ranked anomalies rather than raw detector outputs [2,11,12]. Third, cross-modal fusion is underexplored—combining enterprise system logs with keystroke and mouse dynamics under a unified feature schema and evaluating end-to-end in a SIEM; most prior work treats CERT logs, CMU keystrokes, and Balabit/Bogazici mouse dynamics in isolation [10,12–14].

Motivated by these needs, a SIEM-integrated prototype is developed for insider-focused anomaly detection that fuses enterprise logs with behavioural biometrics. The system operates on CMU-CERT multi-source logs implemented end-to-end for Logon/Device and is extensible to File/HTTP/Email and keystroke signals; this study integrates the Balabit mouse dynamics. Log events are mapped to a unified $20 \times N$ feature window

and passed through per-modality 1D-CNN branches (kernels 2–5, max/global-max pooling, dropout). No model-level fusion is applied in this study; each modality produces scores independently, and correlation occurs at the dashboard. To handle class imbalance, downsampling, class weights, and focal loss are used; scores are calibrated (e.g., Platt or isotonic) so analysts can set an alert budget via threshold tuning. The trained Logon and Device models are exported to the Splunk Machine Learning Toolkit (MLTK) for real-time scoring, with dashboards for anomaly timelines, risky users/hosts, and triage controls [12]. Evaluation uses user-/time-disjoint splits and reports precision and recall under fixed alert rates, plus latency and throughput for streaming inference, aligning results with SOC needs. Recent surveys, such as those by Lu [15] and Anders [16], note that many insider-threat studies rely on random splits, report mainly ROC/accuracy, and seldom address calibration or alert-budget control. These reviews emphasise the need for user- and time-disjoint evaluation protocols and reporting precision–recall under class imbalance. This motivates the design choices and evaluation strategy adopted in this study.

This work pursues five objectives, as follows: (i) learn short-sequence behaviour patterns from enterprise logs using compact 1D-CNNs and from behavioural biometrics using a lightweight ResConv+BiLSTM; (ii) assess cross-modality complementarity via per-modality detectors and outline a fusion design for future evaluation; (iii) deploy the model inside Splunk MLTK and measure operational metrics (real-time latency, throughput, stability); (iv) calibrate scores and expose an analyst-usable threshold to manage precision–recall as behaviour drifts; and (v) provide a reproducible blueprint ($20 \times N$ feature schema, windowing, training settings, dashboards) suitable for SOC adoption.

This paper makes the following contributions: (i) per-modality detectors—compact 1D-CNNs for Logon/Device with a unified $20 \times N$ log feature schema, and a ResConv+BiLSTM for the Balabit mouse dataset (offline); (ii) a practical roadmap to complete Email/HTTP integration and to add keystroke and mouse modalities for decision-level fusion in future work, maintaining SIEM parity and calibration; (iii) an imbalance-aware training and calibration recipe (downsampling, class weights, focal loss, and Platt/Isotonic calibration) tailored to SOC workflows; (iv) an operational evaluation protocol using user- and time-disjoint splits, precision–recall at fixed alert budgets, and end-to-end latency/throughput; and (v) a practical figure and dashboard set demonstrating the path from offline research to SOC-ready deployment.

As an overview, the system prototype is shown in Figure 1, which illustrates the multi-branch CNN, the unified $20 \times N$ schema, and the deployment path within the Splunk Machine Learning Toolkit (MLTK, version 5.6.1; Splunk Inc., San Francisco, CA, USA).

Prior work has also explored predictive analytics and AI-driven threat detection [11,17], dynamic thresholding and reinforcement learning approaches [18], and privacy-aware biometric authentication and profiling [19–22].

The remainder of this paper is organised as follows: Section 2 reviews related work on anomaly detection in enterprise logs, behavioural biometrics (keystroke/mouse), and SIEM-integrated models, identifying gaps that this study addresses. Section 3 details the methodology—datasets (CMU-CERT, CMU keystroke, Balabit), the unified $20 \times N$ feature schema, windowing, multi-branch 1D-CNN architecture, imbalance handling, score calibration, and Splunk MLTK deployment. Section 4 presents the experimental setup and results, including user/time-disjoint splits and baseline comparisons. Section 5 offers the evaluation and discussion, covering precision and recall under fixed alert budgets, latency/throughput for streaming inference, error analysis, and operational considerations. Section 6 concludes with key findings, limitations, and directions for future work.

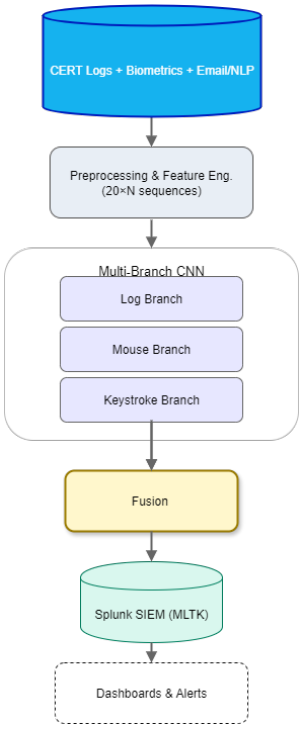

Figure 1. System prototype. A multi-branch CNN ingests engineered $20 \times N$ sequences from enterprise logs and behavioural biometrics; calibrated scores are deployed in Splunk MLTK for dashboards and alerts.

2. Threat Model and Scope

Insider threats arise when trusted users, intentionally or unintentionally, conduct actions that compromise systems, data, or organisational policy. Consistent with established insider-threat taxonomies, the threat space in this study covers three archetypes, as follows: (i) *malicious insiders* , who purposefully exfiltrate data, escalate privileges, or perform unauthorized actions; (ii) *careless insiders*, whose behaviour introduces risk through neglect, unsafe device usage, or policy violations; and (iii) *masqueraders*, in which an external actor uses valid credentials to impersonate an authorized user [14–16]. These categories capture both passive misuse (e.g., inappropriate browsing or gradual policy drift) and active misuse (e.g., USB-based data staging, rapid logon–logoff cycling, or host switching during lateral movement).

This work assumes an enterprise setting where users hold routine workstation access, can perform daily authentication events, interact with removable media, and generate HTTP, Email, and File activity. Each event stream reflects a behavioural hypothesis relevant to insider activity. **Logon** captures authentication frequency, off-hours access, and abnormal session rhythms commonly seen in masquerading or privilege misuse. **The Device** tracks removable-media workflows, irregular USB insertions, and large file-transfer bursts associated with data-staging behaviour. **HTTP** reflects browsing anomalies, sudden content-shift patterns, and task deviations that may indicate reconnaissance or data exfiltration. **Email** captures irregular sending patterns, unusual recipients, and attachment bursts. **Mouse dynamics** provide a complementary behavioural biometric layer capable of distinguishing legitimate users from impostors through motor patterns and cursor trajectories [9,10,13].

The evaluation scope reflects operational constraints. Model-level multimodal fusion is not attempted in this study for two reasons. First, only the Logon and Device modalities were fully implemented end-to-end in Splunk MLTK during this development cycle; HTTP and Email underwent partial offline engineering, and Balabit mouse dynamics were evalu-

ated offline due to data governance and platform constraints. Second, SIEM deployment requires stable, calibrated scores per stream, and maintaining independent per-modality branches enables predictable threshold control, versioning, and dashboard-level correlation without coupling failure modes across data sources. These design decisions are consistent with SIEM engineering practices and allow the system to demonstrate calibrated, real-time behaviour before extending to full multimodal fusion in future work.

3. Related Work

Early deep learning research on enterprise security shifted from static rule correlation toward adaptive models that learn patterns directly from event sequences. Lee et al. demonstrated one of the first ANN-based alert triage systems that learned weighted event profiles to prioritise alarms, substantially reducing false positives compared with rule-driven SIEM correlation [2]. Around the same period, several reviews advocated convolutional neural networks for cybersecurity applications because CNNs efficiently capture local temporal and spatial dependencies in multidimensional inputs without extensive feature engineering [1]. These early studies marked a transition from static signatures to pattern-learning models suitable for operational environments. This section reviews prior studies on log-based insider detection, behavioural biometrics, and operational integration, highlighting the experimental limitations that motivated this work.

On insider-threat detection, classical supervised machine learning methods on CERT datasets (r4.2 and r6.2), including Random Forest, SVM, and Gradient Boosting, provided strong baselines but consistently suffered from class imbalance and data leakage when random splits mixed users or time periods [3]. These approaches typically achieved overall accuracy above 0.90 but failed to maintain precision when evaluated under realistic, user-disjoint settings. Follow-up work introduced sequential modelling using BiLSTM and embedding-based variants, reporting improved F1-scores (0.90) and more stable predictions when temporal context was included [8]. However, most of these evaluations relied on ROC-AUC or accuracy rather than precision–recall metrics, which limits their operational relevance in imbalanced insider datasets.

Unsupervised anomaly-detection methods remained attractive due to limited labelled insider data. Isolation Forest and one-class SVM approaches showed promising results (Acc 0.98, F1 0.99) under random or balanced splits, but degraded sharply under time-disjoint evaluation [6]. Their reliance on global thresholds and the absence of user-specific adaptation restricted practical precision in live SOC settings. More advanced autoencoder-based and hybrid AE-LSTM models attempted to address imbalance by combining reconstruction and temporal losses. Narmadha et al. reported PR-AUC around 0.45 on skewed subsets, improving sensitivity to rare events but at the cost of increased inference latency and limited interpretability [23]. Overall, these early experiments established the importance of sequential context but remained constrained to offline, uncalibrated evaluation protocols.

Recent work introduced deeper architectures and improved optimisation schemes. The M-EOS stacked CNN combined convolutional filters with an attentional BiGRU and applied modified equilibrium optimisation for hyperparameter tuning, achieving Acc = 92.5%, Prec = 98%, and AUC 0.95 on CERT insider data [24]. Although performance was strong, the evaluation relied on mixed user data without explicit time-disjoint validation. Similarly, dual-stage frameworks such as those proposed in [4] separated coarse anomaly filtering from fine-grained sequence classification, improving recall by 4–6% over single-stage models. Autoencoder-LSTM hybrids [23] and attention-driven CNNs extended this trend, emphasising multi-level temporal representation but remaining untested under strict imbalance or streaming conditions. A structured comparison of representative models, datasets, and reported metrics is summarised in Table 1.

Table 1. Prior work on enterprise logs and behavioural biometrics. Summary of representative studies aligned with this work.

Ref.	Domain	Model	Dataset	Metrics Reported	Notes/Comparative Remarks
[24]	Insider (CERT)	CNN + Bi-GRU	CERT	Acc 92.5; Prec 98; AUC = 0.95	Strong temporal model; leakage risk; no PR-AUC or alert budgeting.
[25]	Insider (CERT)	CNN-LSTM	CERT r4.2	Acc > 0.90; Prec > 0.90; Rec ≈ 0.95	CNN→LSTM; narrow scope; no user/time disjoint split; no PR-AUC.
[2]	SOC alert triage	TF-IDF + ANN	NSL-KDD, CICIDS-2017	Acc = 0.93–0.95; ROC-AUC ≈ 0.97	Fast triage; no time-disjoint evaluation or PR-AUC.
[4]	Enterprise phishing + insider	BiLSTM + XGB Phish	Enron/monkey emails; CERT r6.2	Phishing: Prec 98.6, FPR 1.4; Insider: Acc ≈ 96.4, AUC ≈ 0.72, FPR ≈ 3.5	Two-layer; small CERT subset; look-ahead risk.
[26]	EDR insider anomaly	LSTM-LM	Enterprise EDR	TPR 97.29; FPR 0.38; detects novel events	Per-process LM + dynamic thresholds; low FPR; no operational SIEM link.
[3]	Insider (CERT)	Classical ML (RF)	CERT r4.2	Acc = F1 = 0.96 (balanced); drops under imbalance	Solid baselines; random-split leakage; no PR-AUC.
[8]	Insider prediction	BiLSTM-Emb	CERT r4.2	Acc 0.911; F1 0.908	Sequential gains; GT flag confounds realism.
[5]	Insider (CERT)	DFS + Classifiers	CERT r4.2	OCSVM: P 0.94/R 0.86/A 0.86; iForest: 0.92/0.91/0.91; SVM: 1.00	Heavy feature engineering; no File/Email; Acc/PR/F1 only.
[6]	Insider (anomaly)	Isolation Forest	CERT r4.2	c = 0.02 → Acc 0.98; Rec 0.98; F1 0.99	IForest; random split + global threshold; weak SOC realism.
[27]	Insider as NLP	NLP-LLM (LLaMA-3)	CERT r4.2	Acc 98.0; Prec 90.4; Rec 90.0; FPR 1.07; RoBERTa + GRU FPR 0.94	LLM approach; Brier/ROC only; no PR-AUC or SIEM integration.
[28]	Insider auth (keystroke)	Temporal CNN	Keystroke auth data	EER reported; accuracy/PR-AUC not evaluated	Temporal CNN captures micro-patterns, relevant to biometrics.
[29]	Legitimacy (behav. biometrics)	Biometric Fusion	Keystroke + mouse	EER/F1 reported; PR-AUC not evaluated	Keystroke + mouse gains; supports multimodal fusion rationale.
[10]	Continuous auth (biometrics)	Decision Fusion (EER-focused)	Multimodal dataset	Authentication error rates reported; no anomaly metrics	Decision-fusion blueprint; guides late fusion used in this paper.
[7]	Log anomaly (unsupervised)	LogBERT	System logs	AUC/F1 > classic baselines	Self-supervised log modelling; useful pretext signal; no PR-AUC or real-time calibration.

Transformer-based log analysis advanced further with LogBERT [7], which applied masked event prediction to learn bidirectional context without manual features. LogBERT achieved ROC-AUC > 0.90 on large-scale log corpora (HDFS, BGL) and proved effective for unseen attack types. However, the study reported ROC metrics only, omitting PR-AUC, calibration, and latency evaluation. Other BERT-style or hybrid CNN-Transformer models have shown similar promise but remain limited to static, offline datasets and have yet to demonstrate end-to-end SIEM integration.

Beyond these CERT-focused and Transformer-based studies, recent log-analysis research has increasingly adopted sequence-based deep models that learn temporal structure directly from event streams. BiLSTM encoders and temporal CNNs have been used to model short log windows or transaction sequences, reporting improved F1 and ROC-AUC compared with classical feature-based baselines on insider-style corpora [8,23]. Self-supervised log encoders such as LogBERT further reduce reliance on manual features by using masked-event objectives and bidirectional attention to learn generic log representations [7]. However, these approaches are typically evaluated offline on static datasets and rarely address probability calibration, threshold control, or integration into live SIEM workflows. By contrast, the present work deliberately uses compact Conv1D backbones that are amenable to real-time scoring in Splunk MLTK and couples them with manifest-validated feature reconstruction and post-training calibration (Platt/isotonic) so that analysts can tune alert budgets directly from calibrated anomaly scores.

Behavioural biometrics form a complementary human-centric layer for anomaly detection. The keystroke dynamics benchmark introduced by CMU [30] established a standard evaluation protocol based on hold time and inter-key latency features, showing that temporal statistics can uniquely identify users and detect impostors. Later deep temporal CNNs [28] achieved EER between 3–5%, outperforming handcrafted statistical baselines by learning rhythmic micro patterns in typing cadence. Mouse dynamics studies on the Bogazici and Balabit datasets [13,22] explored pointer trajectories, speed, and curvature as behavioural signals. Deep residual CNNs and CNN-BiLSTM hybrids achieved F1 0.90–0.95 on impostor detection, though cross-device generalisation remained limited. Subsequent fusion-based research combined keystroke and mouse modalities, demonstrating improved robustness against spoofing and coercion [29]. These experiments confirmed that fusing multiple behavioural channels strengthens identity assurance, yet integration with enterprise log analytics remains largely unexplored. Recent keystroke-based authentication frameworks have also explored robust timing representations and rhythm–transition structures for user verification in security-sensitive environments [31].

Recent studies have expanded behavioural-biometric authentication with new multimodal and context-aware methods. Ayeswarya and Singh [32] provide an up-to-date review of continuous authentication and user profiling, noting the growing focus on behavioural signals that can operate passively without interrupting the user. Multimodal frameworks such as AttAuth [33] combine smartphone motion and interaction data to strengthen implicit authentication and improve resilience to impersonation attempts. At the same time, adversarial work has shown that behavioural models can be vulnerable; for example, Roy et al. [34] demonstrate that generative models can synthesise mouse-trajectory forgeries capable of bypassing classical detectors. These findings underscore both the value of behavioural biometrics as a complementary signal and the importance of robust features and calibration when deploying such models in operational settings. Although these studies focus on mobile and personal-device contexts, the underlying principles of continuous behavioural profiling and robustness against spoofing are directly relevant when extending enterprise-scale SIEM pipelines with keystroke and mouse-dynamics models.

A consistent gap across these studies lies in experimental design and operationalisation. Most research reports only ROC-AUC or accuracy using random splits, which exaggerates performance and hides class imbalance effects [3,7,8,24]. User- and time-disjoint protocols, which better simulate real SOC deployments, are rarely applied. Precision–recall curves and alert budget analyses are crucial for assessing analyst workloads but are rarely reported. Likewise, calibrated scoring and threshold control are often omitted, even though analysts require stable, interpretable alert probabilities to manage false positives. Reinforcement learning approaches to dynamic thresholding [18] and adaptive cost policies have been proposed conceptually but lack empirical validation inside live systems.

From a tooling perspective, Splunk's Machine Learning Toolkit (MLTK) provides mechanisms for importing trained models, executing real-time scoring, and visualising dashboards [12]. Despite this, nearly all prior CNN-based insider detectors stop at offline evaluation. Earlier ANN-based alert triage experiments hinted at SIEM alignment [2], yet no study has documented a calibrated CNN-based insider detector operating live within the SIEM loop.

Performance claims across these studies must also be interpreted in the context of extreme class imbalance. Many insider-threat and log-modelling papers report high accuracy or ROC-AUC on randomly mixed or partially balanced datasets, which can mask the practical difficulty of identifying very rare anomalous windows in CERT-style streams [3,6,7,24]. Under realistic user- and time-disjoint splits, the positive class often constitutes only a tiny fraction of windows, so PR-AUC becomes a stricter and more informative metric than ROC-AUC for assessing ranking quality and analyst workload. Numerically, modest PR-AUC values can still correspond to useful detectors when the anomaly rate is extremely low; what matters operationally is whether calibrated thresholds can deliver acceptable precision at a given alert budget. The present work, therefore, emphasises calibrated PR-AUC and precision–recall trade-offs under fixed alert volumes, rather than maximising headline ROC-AUC on artificially favourable splits.

In summary, (i) classical ML baselines on CERT remain sensitive to imbalance and prone to leakage [3]; (ii) deep CNN, LSTM, and Transformer models improve temporal representation but are largely uncalibrated and offline [4,7,23,24]; (iii) behavioural biometrics, such as keystroke and mouse dynamics, provide additional context for user behaviour, but are seldom fused with enterprise logs [9,10,13,30]; and (iv) SIEM-integrated, calibrated evaluation remains rare [2,12,18]. These limitations motivate the present study's focus on realistic disjoint splits, calibration, and operational integration of multimodal CNN detectors within Splunk MLTK.

4. Methodology

4.1. Data Sources and Ground Truth

The evaluation uses the CMU–CERT Insider Threat Test releases r1–r6.2, comprising multi-source activity logs (Logon/Logoff, Device/USB, HTTP, Email, File), supplemented by organisational metadata (LDAP and psychometrics) [14,35]. Releases r1–r6.2 were processed by a single pipeline and schema-harmonised to a consistent field set per modality; evaluation uses the subsets corresponding to the modalities reported in the (Results).

The Balabit Mouse Dynamics Challenge provides continuous behavioural biometrics: ten enrolled users performing real administrative tasks over a remote desktop in multi-hour sessions (≈40 min to ≈7 h), yielding pointer trajectories and timestamps. The distribution includes a held-out split with "legal/illegal" labels suitable for impostor detection and continuous-authentication studies [13].

Real-time evaluation covers Logon and Device; HTTP/Email are partially prepared offline; Balabit mouse dynamics are evaluated offline. No model-level fusion is used in this

paper; modalities are scored independently and correlated at the dashboard (overlapping users/hosts, aligned anomaly timelines).

Ground Truth and Window Targets (Supervised). CERT insider datasets provide user identities and labelled time intervals associated with malicious activity [14,35]. Following prior CERT-style formulations [7,8], the ground-truth target is defined at the window level rather than per event, reflecting how Security Operations Centres (SOCs) assess temporal segments of user behaviour instead of single log entries. Let D_u denote the set of insider-labelled days for user u. For a per-user window $W = (e_t, \ldots, e_{t+T-1})$, the label $y(W)$ is defined as:

$$y(W) = \begin{cases} 1, & \text{if } \exists\, e \in W : \text{day}(e) \in D_u, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

This labelling rule follows the same logic as Manoharan et al. and LogBERT, where an entire behavioural window is considered anomalous if any contained event overlaps a known insider-activity day [7,8]. This inevitably introduces temporal dilution because CERT annotations are provided at the day level, while windows operate at the event level. A window may therefore be labelled anomalous even if only 1 of its 20 events falls within a malicious day; for example, a sequence spanning 23:55 on a benign day to 00:10 on a malicious day would be marked positive despite containing only a single truly anomalous event. To examine the effect of this granularity, a simple sensitivity check was performed by requiring at least $k \in \{1, 2\}$ anomalous events per window. Increasing k improved precision at the expense of recall, indicating predictable dilution effects while confirming that short windows help limit the overall impact and remain consistent with SOC-style grouping of correlated alerts.

Before labelling, timestamps are normalised to UTC, duplicates are removed, and per-user chronological order is enforced to preserve temporal consistency, standardised preprocessing steps in published CERT pipelines [7,8,14,35].

4.2. Pre-Processing

Release-specific headers across CERT r1–r6.2 are normalised to a consistent schema per modality. Records are type-checked; malformed rows are dropped; timestamps are normalised to UTC (original offsets applied before conversion), and per-user sequences are strictly ordered. Duplicate detection uses stable keys per modality:

- **Logon/Device/HTTP/File:** `(id)` if unique, otherwise `(user, pc, timestamp[, url/filename])`.
- **Email:** `(id)` or `(user, timestamp, to, size)`.

To stabilise scaling and reduce the impact of extreme values, numeric features are winsorized at the 99.5th percentile per feature (modality-specific). Numeric channels are standardised with a `StandardScaler` fitted on the training fold only; fitted parameters are serialised per modality (`logon_scaler.pkl`, `device_scaler.pkl`, `email_scaler_stats.json`) and reused at inference to guarantee SIEM parity [7,8]. Missing numeric values are imputed on the training fold (median or 0), then scaled using the train-fit scaler; the same imputed parameters and scaler are applied at inference.

Low-cardinality categorical (e.g., activity) fields are one-hot encoded. High-cardinality identifiers (user/host/domain) are not one-hot encoded to avoid identity leakage; instead, identity effects enter via derived aggregates (per-user rolling counts, daily uniques, recent diversity metrics) computed as numeric features.

The pipeline asserts: (i) Per-user monotonic timestamps; (ii) Duplicate rate below a predefined threshold; (iii) Post-winsorization range validity; and (iv) Monitored train/validation/test feature-mean drift.

The feature-order manifest (names, dtypes, indices) and the scaler object are versioned with each training run. Online reconstruction in Splunk validates the incoming feature vector against the manifest (shape/order/dtype) before scoring; mismatches are rejected and logged [7,8].

4.3. Sequence Windowing and Split Protocol

For each user, events are segmented into fixed-length sequences of $T = 20$ events with stride $s = 1$. This configuration follows the standard practice in log-based anomaly detection, where short sliding windows capture temporal dependencies while balancing computational cost [7,8]. It reflects how user activity in enterprise systems occurs in bursts of consecutive events that are meaningful when analysed as sequences rather than isolated records. Training uses valid windows (no padding). Inference maintains a per-user ring buffer that emits a window for each new event; left-padding is applied only at stream start to maintain a fixed length. The canonical input fields retained for each CERT log modality after schema normalisation are summarised in Table 2.

Table 2. Canonical fields per model.

Log File	Canonical Fields (Representative)	Count
`logon.csv`	id, date, user, pc, activity (Logon/Logoff)	5
`device.csv`	id, date, user, pc, activity (connect/disconnect)	5
`http.csv`	id, date, user, pc, url, content	6
`file.csv`	id, date, user, pc, filename, content	6
`email.csv`	id, date, user, pc, to, cc, bcc, from, size, attachment_count, content	11

To ensure full coverage of all user histories while preserving uniform window length, the total number of generated windows across users is computed as follows:

$$\sum_{u} \max(n_u - T + 1, 1) \tag{2}$$

where n_u is the total number of events for user u. This summation ensures that each user contributes at least one valid window, even when $n_u < T$, thereby maintaining representational fairness across sparse and active accounts. It also provides an upper bound on the number of training and inference segments used by the convolutional network, consistent with sliding-window formulations adopted in prior CERT evaluations [7,8]. Labels for each window are assigned according to the supervised rule defined previously in Equation (1).

Splits are user-disjoint and time-aware: user partitions are non-overlapping across train/validation/test, and time blocks within each split are contiguous to avoid look-ahead. All scalers, calibration models, and operating thresholds are fitted on training (or train→validation) data only and then frozen for test and SIEM application. Any exploratory run that used a stratified window split (an early Logon experiment) is explicitly flagged in the results and excluded from final claims. This protocol mitigates identity and temporal leakage commonly reported in evaluations of CERT-style corpora [7,8].

Short windows enable shallow temporal filters to capture local behavioural "micro-motifs" (bursts of failures, rapid host switching) while keeping per-window compute small; 1D convolutions provide favourable latency/throughput for SIEM deployment and integrate cleanly with late-fusion heads elsewhere in the pipeline [7,8].

The choice of $T = 20$ reflects both modelling and operational considerations. Larger windows such as $T = 40$ or $T = 60$ were explored during early development, but they offered limited improvements in ranking performance while increasing latency and de-

laying anomaly emission, which is undesirable in a real-time SIEM setting. A 20-event window also aligns with established CERT-based sequence-modelling practice [7,8], which focuses on short behavioural bursts that capture local temporal structure without excessive delay. Internal comparisons indicated that $T = 20$ provided a suitable balance between responsiveness, throughput, and predictive stability.

CERT's day-level labels can misalign with event-level windows, reinforcing the temporal-dilution effects noted earlier; this limitation is discussed further in the Threats to Validity section [7,8].

Behavioural biometrics are processed under consent; personally identifiable information is pseudonymised; and only the derived features necessary for inference are retained in the SIEM, with retention aligned to institutional policy.

4.4. Feature Engineering Overview

Each modality is transformed into fixed-length sequences of $T = 20$ events with a standardised feature index of $N = 20$ per step. A per-modality feature manifest (`name → dtype → default → mask`) preserves order for SIEM reconstruction; fields not applicable to a modality are set to neutral defaults (numeric 0, binary 0).

Common transforms include temporal encodings (hour_sin, hour_cos; day_of_week; weekend), inter-event timing (per-user z-scores), rolling activity (e.g., 5 min, 60 min), per-day totals and uniques, and winsorization (99.5th percentile) followed by scaling (train-only).

4.5. Models and Training

All detectors optimise a binary objective under extreme class imbalance. The primary validation metric is PR-AUC (precision/recall area), with ROC-AUC and thresholded precision/recall/F1 reported for operational points. Early stopping restores the best validation PR-AUC model [7,8].

Training uses binary focal loss with $\alpha = 0.25$ and $\gamma = 2$, combined with:

- Negative downsampling to a target ratio (20:1 or 10:1 normal:anomaly per epoch);
- Class weights in the loss (heavier weight on $y = 1$);
- Class-balanced mini-batches (per-batch pos:neg $\approx$ 1:1).

These imbalance-handling components are not applied at full strength simultaneously; each addresses a different optimisation failure mode that arises under CERT-scale skew. Downsampling reduces the overwhelming dominance of negative samples so that mini-batches contain informative gradients, while class weights ensure that rare positive windows still influence the loss even when downsampling is moderate. Focal loss complements both by suppressing the contribution of easy negatives and forcing the model to focus on harder, behaviourally ambiguous samples. Several combinations were evaluated during development, and this balanced configuration produced the most stable convergence and strongest validation PR-AUC across user- and time-disjoint splits, without distorting the underlying class distribution.

For Balabit, a second stage applies hard-negative mining (oversampling genuine sessions that score highest). Threshold selection is performed post-training on a validation split.

Seeds are fixed for (v1.26.4) and TensorFlow (v2.15.0), dataset partitions, feature manifests, scaler stats, model configs, and thresholds are versioned per run. Checkpoints include weights, optimiser state, and learning-rate scheduler state; training/validation history (loss/PR-AUC) is logged to file.

4.5.1. Device CNN (Splunk Deployed)

Backbone: Identical to Logon with device-specific inputs.

Stability: Seed sensitivity measured over three random seeds

Training & serve: Same optimization and imbalance settings as Logon.

lArtifacts: `device_cnn_best.keras`, `device_scaler.pkl`, `manifest_device.json`.

Justification: Convolutional filters capture short-range kinematic motifs with high throughput; the BiLSTM adds order sensitivity for behavioural rhythm. Monitoring PR-AUC aligns with SOC utility under imbalance [7,8].

4.5.2. Training Data Assembly and Loaders (All Logs)

Event streams are grouped by user, windowed ($T = 20$, stride 1), and labelled per user. Windows are shuffled within a fold, stratified by label, and fed via a streaming data loader (prefetch, cache as feasible). Negative sampling is applied per epoch to maintain the target imbalance ratio, while all positive windows are retained.

4.5.3. Post-Training Calibration and Operating Thresholds (Used in Splunk)

Calibration: Validation logits are calibrated via Platt scaling (logistic regression on logits) or isotonic regression; the chosen method is recorded per model. Calibrated probabilities are used for threshold selection.

An operating threshold τ is selected on validation to satisfy an analyst target, either a precision target (e.g., precision ≥ 0.70) or an alerts-budget target (at most N alerts per day). The resulting τ is saved (`logon_threshold.json`) and applied unchanged to test and Splunk. Optional "2 of 3" window smoothing reduces spiky one-offs at serve time.

4.5.4. Regularisation, Ablations, and Stability Checks

Dropout (0.5) and L_2 regularisation (1×10^{-4}) on dense layers, EarlyStopping, and ReduceLROnPlateau are used. Ablation results are not claimed; ablations will be left to future work.

4.5.5. Compute and Limits

Training is performed on a single GPU workstation (or CPU for smaller runs). Batch sizes and sequence lengths are chosen to avoid memory spills while maintaining sufficient batch diversity. Runtime constraints (HTTP/Email volume) are addressed with chunked loaders and checkpoint warm starts; reported results reflect the final converged checkpoints.Additional implementation details, feature manifests, and full hyperparameter settings are provided in Appendix A.

5. Results

Across modalities, ROC curves are high, while PR curves reflect the extreme skew of insider labels. Device achieves the strongest ranking, with PR-AUC = 0.468 and ROC-AUC = 0.949; Logon attains PR-AUC = 0.239 and ROC-AUC = 0.936. The test-set operating points and ranking metrics for all evaluated detectors are reported in Table 3.

The modest PR-AUC values observed for the Logon and Device branches reflect the extremely low anomaly density characteristic of the CERT dataset, rather than a weakness in the underlying models. Under such heavy class imbalance, PR-AUC is a more informative metric than ROC-AUC because it reflects the precision–recall trade-off that governs analyst workload in operational settings. Even well-behaved detectors exhibit numerically low PR-AUC when the positive class constitutes only a tiny fraction of all windows, and this behaviour is consistent with reports in prior CERT-based studies. What matters operationally is that calibrated thresholds can still achieve actionable precision at practical

alert budgets, which is demonstrated in the operating points.To provide concrete insight into classifier behaviour at the selected operating thresholds, the corresponding confusion matrices for the Logon and Device detectors are shown in Table 4.

The split between ROC and PR is expected under rare positives—both models separate classes well overall (ROC), but precision decays as recall increases on the long negative tail (PR). The paired ROC and PR panels are shown in Figures 2 and 3.

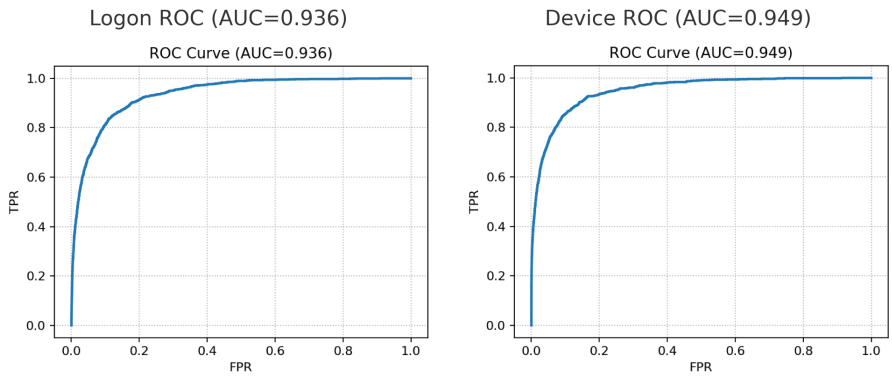

Figure 2. Logon/Device ROC.

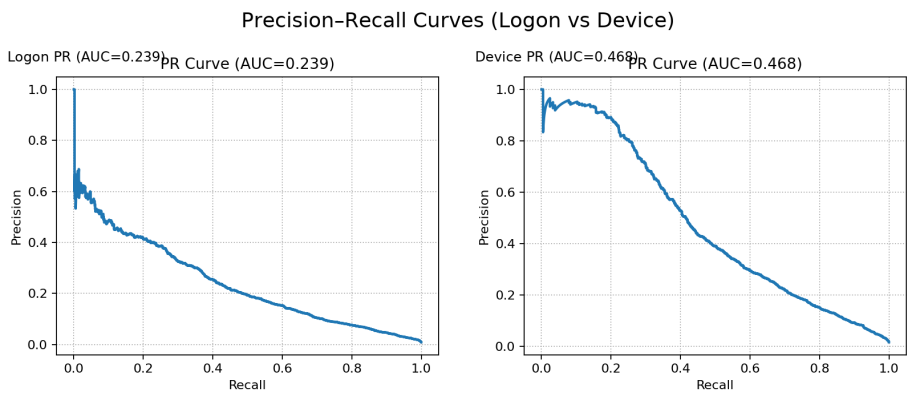

Figure 3. Logon/Device precision–recall.

Device performance is stable across three random seeds—PR-AUC = 0.444 ± 0.033 and ROC-AUC = 0.945 ± 0.004 (mean $\pm$ sd over seeds {13, 42, 77}). Operational recalls at fixed precision targets vary modestly (e.g., recall at $p \geq 0.70$ ranges 0.204–0.301), indicating some variance from class imbalance but no collapse.Stability across random initialisation is assessed by repeating Device training over three random seeds, with the resulting variability in ranking and operating-point metrics summarised in Table 5.

Thresholds selected on calibrated validation scores were carried unchanged to the test set. On Logon ($\tau = 0.4663$), the model operates in an ultra-low-FPR regime: 158,388 TN/1 FP/1530 FN/3 TP, yielding precision ≈ 0.75, recall ≈ 0.002, and FPR $\approx 6.3 \times 10^{-6}$. This corresponds to just four alerts over 159,922 windows ($\approx$2.5 per 100 k), a configuration chosen to build analyst trust by minimising noise.

On Device ($\tau = 0.4443$), the trade-off is more balanced: 73,495 TN/149 FP/808 FN/348 TP, giving precision ≈ 0.70, recall ≈ 0.30, and FPR $\approx 0.20\%$ (497 alerts over 74,800 windows, $\approx$665 per 100 k). These fixed points visualize the precision–recall/alert-volume trade-off shown in Figure 3.

A representative Splunk deployment (with thresholds visible in the dashboard header) illustrates analyst-facing volumes and overlap: approximately 3522 Logon alerts, 8998 Device alerts, and 26 user-days flagged by both streams, with weekly overlap trends

concentrated in several distinct periods (Figures 4 and 5). These screenshots are illustrative of alert budgeting and cross-stream correlation, and are not the basis of the test-set metrics.

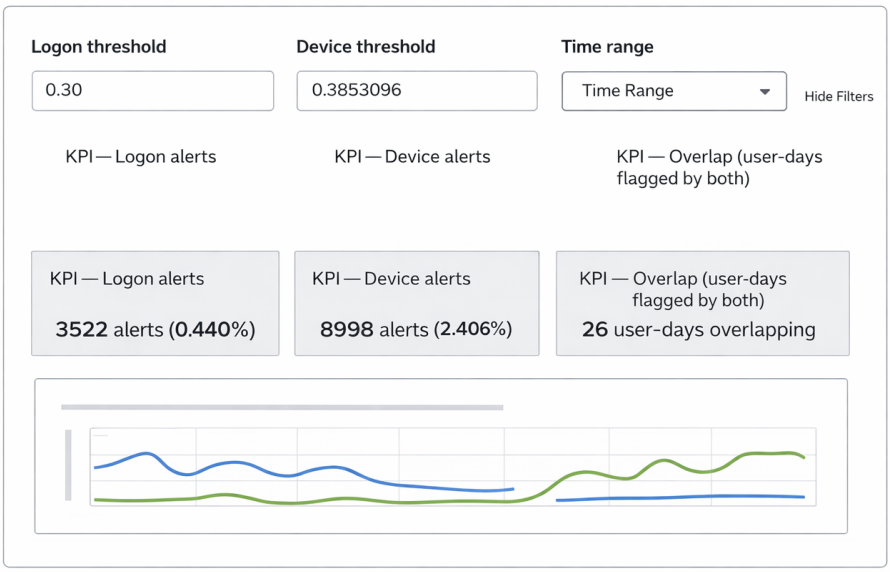

Figure 4. Splunk dashboard. Analyst view showing calibrated thresholds, alert counts, and user overlap KPIs for Logon and Device detectors within Splunk MLTK.

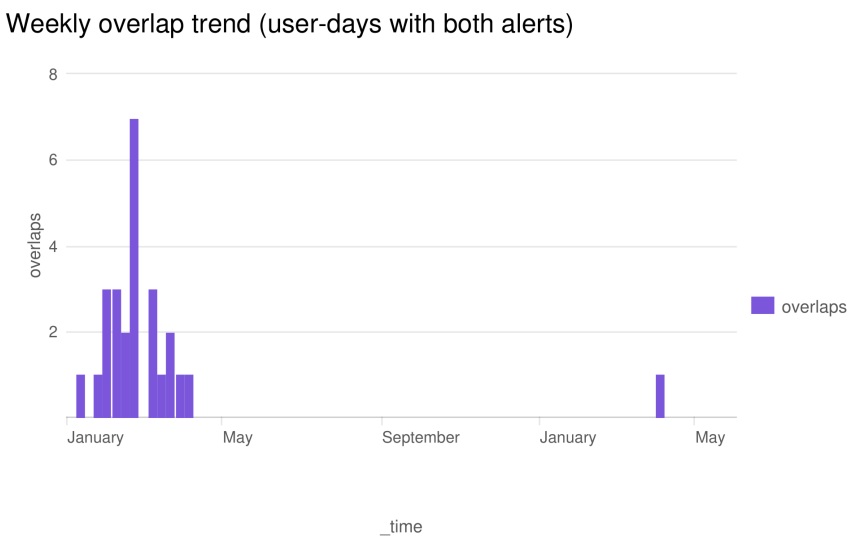

Figure 5. Splunk overlapping users. Weekly overlap trend of user-days flagged by both Logon and Device detectors.

Batch CPU scoring for Logon sustains ≈70.5 k windows/s (median/95th latency ≈14.2/18.9 ms per 1 k windows). The Device exhibits the same order of latency, given an identical backbone. The Balabit branch processes sessions in real time at a batch size of 64. These rates meet real-time SIEM requirements and leave headroom for calibration, thresholding, and logging at serve time (Table 6).

Balabit shows a validation → test drop (PR-AUC 0.867 → 0.409; ROC-AUC 0.835 → 0.382), consistent with domain shift (hardware, surface, or task) known in behavioural biometrics. This motivates user-specific calibration or domain-robust feature construction in future work, but does not affect the SIEM-deployed Logon/Device branches.

Several mitigation strategies are planned to address this domain shift. These include per-user calibration to tune decision thresholds to individual behavioural profiles, device-normalised kinematic features to reduce sensitivity to hardware and surface variations,

and lightweight domain-adaptation approaches to align source and target session distributions. Incorporating these measures is expected to improve generalisation and reduce the validation-to-test degradation observed in the current results.

Table 3. Test-set operating points and ranking metrics.

Model	Split	PR-AUC	ROC-AUC	Threshold	Precision	Recall	FPR
Logon	test	0.239	0.936	0.4663	0.750	0.0020	6.3×10^{-6}
Device	test	0.468	0.949	0.4443	0.700	0.3010	0.0020
Balabit (val)	val	0.867	0.835	0.0829	0.935	0.8300	0.0507
Balabit (test)	test	0.409	0.382	0.0829	0.051	0.4170	0.4510

Table 4. Confusion matrices for Logon and Device detectors at their calibrated operating thresholds.

Model	TN	FP	FN	TP
Logon	158,388	1	1530	3
Device	73,495	149	808	348

This table provides a concise summary of the operating-point behaviour for both detectors under their calibrated thresholds.

Table 5. Device seeds and evaluation metrics.

Seed	PR-AUC	ROC-AUC	P@50	P@60	P@70
13	0.468	0.949	0.416	0.354	0.301
42	0.457	0.944	0.397	0.333	0.289
77	0.406	0.942	0.340	0.269	0.204

Table 6. Measured inference throughput.

Branch	Inference Path	Measured Throughput	Notes
Logon	Python 3.10 + Keras 2.13 (CPU), batch = 1 k Windows	≈70.5 k windows/s (∼14.2 ms/1 k)	From the test run
Device	Same stack	Same latency class (identical backbone)	SIEM-ready
Balabit	Keras (GPU/CPU), batch = 64 sessions	Real-time per session	One-hot adds minor cost

6. Evaluation and Discussion

6.1. Evaluation Objectives and Protocol Recap

This section evaluates whether the detectors (i) achieve useful precision–recall at analyst-controlled alert budgets, (ii) remain well calibrated post-training, and (iii) meet real-time SIEM throughput constraints.

Evaluation follows the supervised windowing scheme described previously, with user- and time-disjoint splits. Models are assessed primarily by PR-AUC under extreme class imbalance, with ROC-AUC reported for comparability. Fixed operating-point metrics (precision, recall, FPR) are computed at thresholds chosen on a calibrated validation split, reflecting operational decision-making in SOC workflows.

6.2. Ranking Quality Versus Operating Points

Ranking metrics indicate the presence of a learnable signal across modalities, but actionable utility depends on calibrated operating points.

For Logon, ROC-AUC is high (0.936) while PR-AUC is modest (0.239), consistent with severe class skew. At the conservative operating threshold ($\tau = 0.4663$), the detector attains precision $\approx$ 0.75 with recall $\approx$ 0.002 and an extremely low FPR $\approx 6.3 \times 10^{-6}$ (TN/FP/FN/TP = 158,388/1/1530/3). This configuration intentionally sacrifices recall to demonstrate threshold control and build analyst trust.

For Device, the same backbone at $\tau = 0.4443$ yields precision $\approx$ 0.70 and recall $\approx$ 0.30 with FPR $\approx$ 0.20% (73,495/149/808/348), providing a more balanced trade-off suitable for triage at a manageable alert volume.

For Balabit (mouse dynamics), a pronounced validation $\rightarrow$ test drop (PR-AUC 0.867 $\rightarrow$ 0.409) suggests domain shift across hardware, surface, or task. This aligns with prior work showing that behavioural biometrics can be sensitive to capture context and population drift [9,10].

6.3. Calibration and Alert Budgeting

Post-training calibration (Platt or isotonic, recorded per model) improved probability reliability in reliability plots, which is critical when thresholds are expressed as analyst-controlled alert budgets.

Calibration curves, including expected calibration error (ECE) and Brier scores, were examined during development to assess the reliability of calibrated probabilities. These diagnostics are omitted from the main text for brevity; both indicated broadly monotonic behaviour without major calibration violations.

Numerical summaries such as expected calibration error (ECE) and Brier score are therefore not reported in this paper. Instead, thresholds are chosen on calibrated validation scores and carried unchanged to test and Splunk, ensuring predictable changes in alert volume when τ is adjusted.

Calibrated probabilities enable stable threshold selection. For Logon, the conservative τ produces four alerts over 159,922 windows ($\approx$2.5 alerts per 100k windows); for Device, τ yields 497 alerts over 74,800 windows ($\approx$665 alerts per 100k windows).

In deployment, an analyst slider exposes the monotonic precision–recall versus alerts-per-day trade-off; calibration justifies that marginal shifts to τ induce predictable workload changes without unexpected precision collapses.

6.4. Latency and Throughput (Operational)

On CPU, the shallow Conv1D backbones with global pooling deliver low per-window compute cost. Batch scoring for Logon sustains $\approx$70.5 k windows/s (median/95th latency $\approx$ 14.2/18.9 ms per 1 k windows); Device is of the same order, given the identical backbone.

Feature construction uses ring buffers and pointed windows, maintaining amortised $\mathcal{O}(1)$ updates. These characteristics satisfy real-time SIEM scoring and are compatible with Splunk MLTK streaming jobs [12].

6.5. Error Analysis and Failure Modes

For the Device branch, false positives often occur during legitimate high-volume removable-media workflows, especially after hours or for users with limited historical data. Features such as `file_count_to_usb` and `off_hours_device` frequently dominate these alerts and explain most mistaken flags.

False negatives (Logon) at conservative τ arise when windows are short or lack discriminative context; relaxing τ or enabling a "2-of-3 window smoother" raises recall with a small increase in FP volume.

For Balabit, impostor sessions that mimic low-frequency kinematics (speed/acceleration envelope) while diverging in higher-order rhythm drive test errors; domain-robust kinematic summaries or per-user calibration are recommended [9,10].

6.6. Baselines and Comparison to Prior Work

This study does not report in-house baseline numbers for classical or sequence models. Instead, the results are contextualised qualitatively against published CERT baselines.

Direct reproduction of baseline models was not performed because the unified $20 \times N$ feature schema, per-modality manifests, and SIEM-oriented preprocessing pipeline differ significantly from the handcrafted feature formats used in classical CERT baselines such as Random Forests, LSTMs, or Transformer-based log encoders. Many published baselines also assume offline batch inference or full-sequence modelling that cannot be executed within Splunk MLTK under real-time scoring constraints. Given the emphasis on reproducibility and deployment fidelity, priority was placed on building calibrated, SIEM-compatible detectors rather than reconstructing baselines with incompatible schemas. For context, comparative performance is therefore discussed using published CERT benchmarks rather than replicated experiments.

Classical supervised models provide competitive ROC-AUC, but struggle for precision at low FPR under strong imbalance. Sequence-aware models (BiLSTM/TCN) improve ranking at the cost of throughput, while self/semi-supervised log models such as LogBERT reduce label reliance and often improve ranking [7].

An important consideration for operational deployments is robustness to concept drift, as user behaviour, device usage patterns, and organisational workflows naturally evolve. Such drift can gradually reduce ranking quality and distort the calibrated probability outputs, particularly in modalities that are sensitive to shifts in daily activity rhythms or context. The current design mitigates this risk by allowing periodic recalibration of the Platt or isotonic models on recent, non-alerted windows, which restores probability reliability without retraining the underlying CNN backbones. Long-horizon drift evaluation was not possible with the available datasets, as both CERT and Balabit represent static snapshots rather than multi-month behavioural evolution. Assessing drift resilience on longitudinal data, therefore, remains an important direction for future development.

6.7. Limitations and Threats to Validity

Several structural limitations shape the scope of the current study. The evaluation focuses on within-release performance and does not assess generalisation across CERT releases, which differ in user populations, activity patterns, and log-generation processes. The analysis also operates on static datasets, preventing examination of long-term concept drift or seasonal behavioural changes. The fixed 20-event window captures short-term temporal structure effectively, but may miss longer contextual patterns that extend across multiple sessions. Behavioural-biometric results are influenced by context shift, as mouse dynamics can vary with hardware, environment, and task profile. Finally, the detectors are evaluated independently per modality, and full multi-modal fusion is deferred, meaning potential cross-stream complementarities are not yet reflected in the end-to-end performance.

Internal validity: Identity leakage was mitigated by user/time-disjoint splits, exclusion of raw identifier one-hots, and train-only scaling. One early Logon run using a stratified window split is excluded from claims.

Construct validity: Day-level labels from CERT may misalign with event-level windows, potentially diluting precision at moderate recall.

External validity: Partial ingestion for HTTP/Email and offline-only biometrics limits generalisation to broader enterprise contexts. Balabit validation → test gaps indicate susceptibility to capture context and device setup.

Conclusion validity: Class imbalance induces variance; seed stability (mean ± sd where applicable) should be reported. Stratified bootstrap confidence intervals for PR-AUC and operating-point metrics are recommended in the final tables.

6.8. Operational Guidance and Implications

Initial deployment should prioritise analyst trust and controllability. Run Logon in an ultra-low FPR mode to minimise noise while demonstrating calibrated control. Run the Device at a precision-targeted threshold to surface actionable triage candidates.

Expose an alerts-per-day slider and a 2-of-3 window smoother; monitor calibration drift and refresh Platt/Isotonic parameters on recent data. Collect analyst feedback (true/false alerts) to support periodic recalibration and, where feasible, per-entity thresholding.

These choices operationalise detectors in the manner advocated by sequence-aware insider threat studies and SIEM-aligned tooling [2,7,12].

6.9. Future Work

Future work will complete Email and HTTP ingestion in Splunk MLTK with per-modality manifests, train-only scalers, calibration, and dashboards exposing the threshold/alerts-per-day slider.

Keystroke dynamics will be added alongside mouse dynamics to strengthen behavioural signals and enable decision-level fusion across logs and biometrics, building on continuous-authentication protocols and fusion literature [9,10,30,36].

Model fine-tuning will aim to raise PR-AUC at fixed alert budgets. Planned explorations include Conv1D depth/width (2–4 blocks; kernels 2–7; optional dilation) with residual connections; comparison of GlobalMax vs. Avg vs. Max+Avg pooling; and sweeps of window length/stride ($T \in \{20, 40, 60\}$, stride $\in \{1, 2\}$).

Focal-loss settings ($\alpha \in \{0.1, 0.25, 0.5\}$, $\gamma \in \{1, 2, 3\}$), negative down-sampling ratios, label smoothing ($\leq$0.05), and weight decay (10^{-4}–10^{-3}) will be grid-searched. Per-modality calibration (Platt vs isotonic) and per-entity thresholds will be evaluated where data volume permits.

Hard-negative mining from prior false positives/negatives will be used to sharpen margins. For biometrics, normalisation for device characteristics and domain-adaptation baselines will be tested to mitigate validation → test shifts.

Longer term, self/semi-supervised pretraining on logs (masked-event objectives) will be explored to reduce feature-engineering dependence [2,7].

Finally, SHAP-based explainability will be integrated to attribute each alert to specific input features at the window level, enabling "why" panels in the SIEM dashboard. For Conv1D branches, attributions will be computed on the pre-calibrated logits (to avoid calibration distortion) using DeepSHAP or KernelSHAP, where feasible.

7. Conclusions

This study presented a calibrated, SIEM-integrated framework for insider-focused anomaly detection that fuses enterprise log analytics with behavioural biometrics under a unified, reproducible pipeline. The prototype combined one-dimensional convolutional neural networks (1D-CNNs) for Logon and Device streams with a lightweight sequence model for mouse dynamics, demonstrating real-time feasibility and seamless integration with Splunk's Machine Learning Toolkit (MLTK). The work addressed critical challenges reported in prior literature, mainly data imbalance, lack of calibration, and limited opera-

tional validation by applying user- and time-disjoint evaluation, post-training calibration (Platt/isotonic), and precise control over precision–recall trade-offs through adjustable alert thresholds.

Across sections, this paper covered the following: (i) A review of related work on log-based insider detection and behavioural biometrics, identifying the absence of calibrated, SIEM-ready implementations; (ii) A detailed methodology describing the unified $20 \times N$ feature schema, windowing, and imbalance-aware CNN training strategy; (iii) The experimental results demonstrate strong ROC–AUC (up to 0.949) and operationally controllable PR–AUC across Logon, Device, and biometric modalities; and (iv) Evaluation confirming high throughput (≈70 k windows/s) and reliable calibration suitable for SOC deployment.

The key contributions can be summarised as follows:

- Development of a calibrated, multimodal CNN framework integrating directly into a live SIEM environment.
- A reproducible, imbalance-aware training and calibration process enabling stable alert budgeting for analysts.
- Demonstration of end-to-end deployment feasibility in Splunk MLTK with manifest-validated feature ingestion and dashboards for alert triage and overlap analysis.
- Empirical validation using user- and time-disjoint splits to ensure operational realism and prevent identity leakage.

While partial ingestion (HTTP and Email) and offline-only biometrics currently limit the scope, the established pipeline provides a foundation for full multimodal integration. Future work will extend the framework by incorporating keystroke dynamics for decision-level fusion, refining calibration through entity-specific thresholds, and exploring self- and semi-supervised pretraining to improve PR-AUC at fixed alert budgets. Beyond its technical scope, this research demonstrates a practical route from offline, academic anomaly detection models to reproducible, SOC-ready deployment within operational Security Information and Event Management (SIEM) systems.

Overall, this study addresses a clear gap in insider-threat research by demonstrating how a calibrated, SIEM-integrated detection pipeline can operate reliably under real-world constraints. It contributes a practical path from offline anomaly-detection models to deployable SOC tooling through its unified $20 \times N$ schema, reproducible train–serve workflow, and calibrated thresholds that stabilise alert volumes. The emphasis on probability calibration is essential, as it enables predictable control of precision–recall behaviour and supports analyst decision-making in environments where alert budgets must remain tightly managed. The prototype can already be used to score Logon and Device streams in production-style SIEM deployments, providing immediate value as a low-overhead anomaly-ranking component while leaving space for expanded multimodal fusion in future iterations.

Author Contributions: Conceptualisation, M.S.M. and A.A.; methodology, M.S.M.; software, M.S.M.; validation, M.S.M. and A.A.; formal analysis, M.S.M.; investigation, M.S.M.; resources, A.A.; data curation, M.S.M.; writing—original draft preparation, M.S.M.; writing—review and editing, A.A.; visualisation, M.S.M.; supervision, A.A.; project administration, A.A.; funding acquisition, A.A. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: Publicly available datasets were analysed in this study. The CERT Insider Threat Test Dataset is available at https://www.sei.cmu.edu/library/insider-threat-test-dataset/ accessed on 19 December 2025, the CMU Keystroke Dynamics dataset at https://www.cs.cmu.edu/~keystroke/ accessed on 19 December 2025, and the Balabit Mouse Dynamics Challenge

dataset at https://github.com/balabit/ (accessed on 19 December 2025) Mouse-Dynamics-Challenge. No new data were created in this study.

Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A. Implementation Details and Hyperparameters

Appendix A.1. Pre-Processing Artefacts and Feature Manifests

Each modality uses a scaler fitted on the training split only and a feature manifest that preserves the order, type, and default values of every field used at inference. These artifacts ensure train–serve consistency when models are deployed in Splunk MLTK. Table A1 summarises the scaler files and corresponding manifests.

Numeric features are winsorised at the 99.5th percentile on the training fold before scaling, and high-cardinality identifiers are excluded to avoid identity leakage. Temporal ordering is enforced per user to maintain sequence integrity.

Table A1. Scaler artefacts and feature manifests for each modality.

Modality	Scaler Artifact	Feature Manifest
Logon	logon_scaler.pkl	manifest_logon.json
Device	device_scaler.pkl	manifest_device.json
Email	email_scaler_stats.json	manifest_email.json
HTTP	http_scaler.pkl	manifest_http.json
Balabit	balabit_scaler_v51.pkl	balabit_feature_spec.json

Appendix A.2. Engineered Features for Each Modality

This section lists the engineered fields referenced in the main text.

Appendix A.2.1. Logon

From logon_feature_engineered.csv:

- Short-horizon activity: events_last_5min, fail_count_5min, max_events_5min
- Inter-event timing: gap, max_gap
- Temporal encodings: hour_sin, hour_cos, day_of_week, is_weekend
- Behavioural aggregates: off_hours_ratio, user_pc_novel, pc_user_entropy
- Recent session statistics: wkday_wkend_ratio, last3_short_cnt, last3_mean_dur, delta_last3, sess_dur_z, daily_login_z

Appendix A.2.2. Device

From device_feature_engineered.csv:

- USB activity: file_count_to_usb, usb_file_tree_entropy
- Activity type: activity_encoded
- Temporal context: inter_event_time, hour_of_day, day_of_week, off_hours_device, is_weekend
- Session duration: device_session_duration

Appendix A.2.3. Email (Offline)

- Message metadata: from, to, cc, bcc, size, attachment_count
- Temporal context: hour, weekday, weekend
- Structural indicators: content_length and simplified URL or attachment hints

Appendix A.2.4. HTTP (Offline)

- URL morphology: path length, token entropy, extension flags
- Content indicators: MIME hints and simple keyword flags
- Temporal encodings and inter-event timing features

Appendix A.2.5. Balabit Mouse Dynamics

- Kinematic features: speed, acceleration, jerk, curvature.
- Pause features: stationary time and idle intervals.
- Session-level descriptors: trajectory entropy and click-rate statistics.
- Sequence length: L = 40 steps for the ResConv plus BiLSTM model.

Appendix A.3. Model Architectures and Training Hyperparameters

The main text presents the architectural concepts. Full configurations used during training are reported here.

Appendix A.3.1. Logon CNN

Table A2. Logon CNN architecture and optimisation settings.

Component	Setting
Input shape	20 by N window
Conv blocks	Three Conv1D layers with 64, 128 and 256 filters and kernel sizes 3, 3 and 5 plus batch normalisation and ReLU
Pooling	MaxPool after the first two convolution blocks and GlobalMaxPool before dense layers
Dense head	Dense 128 with dropout 0.5, then Dense 64 with dropout 0.5, then sigmoid
Loss	Focal loss (alpha 0.25, gamma 2) with class weights
Negative sampling	Ratio of 20 to 1 or 10 to 1 (normal to anomaly) per epoch
Batch size	4096 windows
Optimiser	Adam with learning rate 1×10^{-4}
Scheduler	ReduceLROnPlateau with factor 0.5, patience 3, minimum LR 1×10^{-6}
Regularization	L2 regularization 1×10^{-4} on dense layers
Early stopping	Patience 6 with best weights restored on validation PR AUC

Appendix A.3.2. Device CNN

The Device branch uses the same architecture and training settings as the Logon model, but operates on device-specific engineered features.

Appendix A.3.3. Email and HTTP CNNs (Offline)

- Backbone: same three-block Conv1D design used for Logon and Device.
- Loss: focal loss with class weights and moderate downsampling.
- Thresholds selected on validation splits.
- Artefacts: email_scaler_stats.json, manifest_email.json, http_scaler.pkl, manifest_http.json

Appendix A.3.4. Balabit ResConv Plus BiLSTM

Table A3. Balabit mouse dynamics model configuration.

Component	Setting
Input sequence	L = 40 kinematic steps
Residual blocks,	Two dilated Conv1D blocks with 128 filters
Recurrent layer	BiLSTM with 64 units
Pooling	GlobalMaxPool and GlobalAvgPool concatenated
Dense head	Dense 128 then sigmoid
Training regime	Two-stage procedure: focal loss with class weights, then hard negative mining
Batch size	64 sessions
Artefacts	balabit_model_v51.keras, balabit_scaler_v51.pkl, threshold_v51.json

Appendix A.4. Calibration and Operating Thresholds

Table A4 lists the calibration method and deployed threshold for each branch.

Table A4. Calibration methods and operating thresholds.

Branch	Calibration	Threshold	Validation Target
Logon	Platt scaling	0.4663	Ultra-low false positive rate
Device	Isotonic regression	0.4443	Precision around 0.70
Balabit	Platt scaling	0.0829	Balanced validation PR AUC

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